Michelle Zuo

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EDUCATION

Duke University, Durham, NC — Aug. 2026 – May 2030 (Expected)

Mathematics and Computer Science Major

Coursework: Data Structures and Algorithms, Linear Algebra with Proofs, Advanced Problem Solving Seminar

Activities & Societies: Catalyst, Duke Quantitative Finance Club, HackDuke (Operations Team)

Thomas Jefferson High School for Science and Technology, Alexandria, VA — Aug. 2023 – June 2026

Honors: AIME Qualifier (4x), AP Scholar with Distinction | Weighted GPA: 4.589

Coursework: Research Statistics 3 (R, statistical and probability programming), Artificial Intelligence 1 and 2, Concrete Math, Multivariable Calculus (dual enrollment)

EXPERIENCE & RESEARCH

Data Analyst, Duke Impact Investment Group (DIIG), Durham, NC — Sep. 2026 – Present

- Transforming 9 research questions into a roadmap for ADYN (YC25) by aligning business and scientific goals
- Analyzing 7 datasets (orders, hormone, genotype) with a 5-person team through documented exploratory data analysis and data cleaning
- Building a customizable dashboard demo so ADYN staff can self-serve insights, backed by a metrics data dictionary

Research Intern, NSF-Funded DISES Project, George Mason University (ASSIP) — June 2025 – Nov. 2025 *(part-time)*

- Correlated economic loss with social vulnerability across NOAA's top 10 billion-dollar hazards to guide public disaster-aid decisions
- Processed and analyzed terabyte-scale datasets in Python and GeoDa using statistical modeling and spatial visualization
- Co-authored a multi-hazard disaster loss paper, now under review for journal submission

Research Student, Institute for Mathematics and Democracy (MAA-Sponsored), Remote — June 2025 – Apr. 2026 *(part-time)*

- Modeled Wisconsin election data with Möbius inversion and machine learning to analyze swing-state behavior
- Authored a paper preprinted on arXiv (2510.26867), under review at Rose-Hulman Undergraduate Math Journal
- Selected through a competitive national process, presented findings to all program teams, then extended the work with the professor

Quantum Research Student, Virginia Tech Quantum Lab Partnership — Sep. 2025 – June 2026 *(during high school)*

- Worked to improve ADAPT-VQE, a quantum algorithm for molecular simulation, by leveraging entanglement in Python
- Iterated on algorithm design with a faculty mentor and 2 fellow researchers through bi-weekly research meetings

PROJECTS

FindAQure | *Python, Flask, Grover's Algorithm, HTML* — Mar. 2026

- Built a full-stack app that ranks antibody–antigen docking sites from FASTA input to speed up antibody discovery
- Applied Grover's quantum search for a quadratic speedup, O(N) to $O(N^{1/2})$, over classical docking search
- Validated the pipeline end-to-end on an HIV-1 antigen/antibody pair from RCSB PDB

Online Shoppers Purchase Prediction | *Python, scikit-learn, pandas, Random Forest* — Oct. 2026

- Built with 3 Duke teammates a Random Forest predicting purchases at 0.92 ROC-AUC and 90.5% accuracy, beating the 84% majority-class baseline by 6.5 pts
- Eliminated data leakage across 12,205 sessions by fitting a 64-feature pipeline on training data only; evaluated with ROC and confusion-matrix analysis

TECHNICAL SKILLS

Programming: Python, SQL, R, Java, MATLAB, C++ (beginner), LaTeX

Libraries & Tools: Git, scikit-learn, pandas, Flask, GeoDa

Focus Areas: Algorithms, Machine Learning, Statistical Modeling, Probability, Quantum Computing, Spatial Analysis, Data Visualization

Spoken Languages: Chinese